RAIR and KAIR

Multi-hop question answering with inspectable evidence

A local research prototype and controlled comparison

Student or team _____
College _____
Guide _____

Final project presentation | 27 October 2026

The multi-hop question answering task

In which city was the author of The Glass Orchard born?



The system must preserve both links and show their sources.

An unsupported link should lead to abstention.

RAIR role-based retrieval

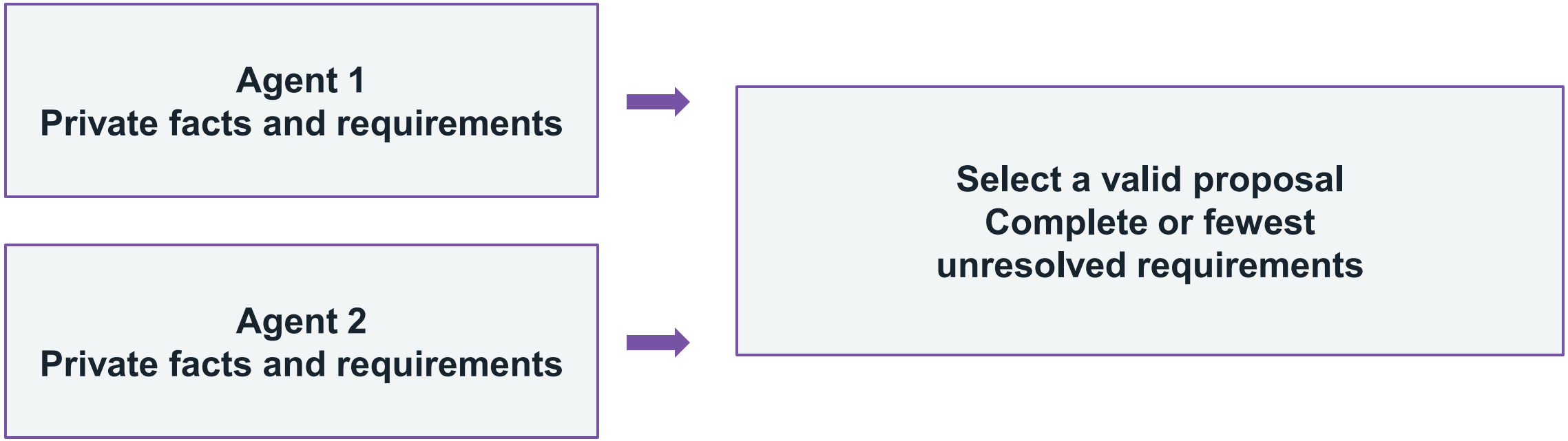


Shared state

Accepted facts, gaps and query history guide the next retrieval.

Short source aliases and required candidate decisions reduce malformed outputs.

KAIR competitive retrieval



Share selected source context while keeping private knowledge separate.

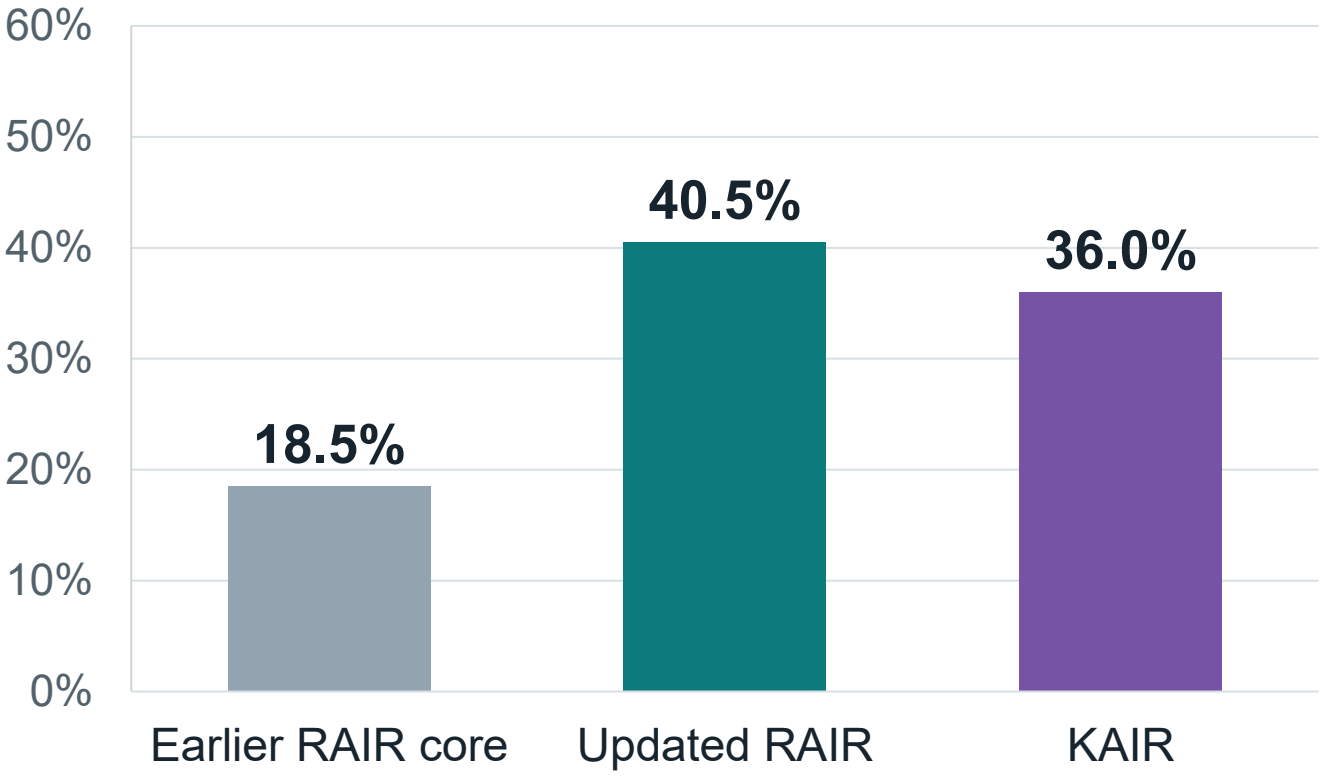
Local execution is serial and bounded.

Fair comparison protocol

Control	Frozen protocol
Question cohort	100 fresh questions after 125 exposed IDs were excluded
Runs	2 seeds and 3 arms give 600 runs, 200 per arm
Model and sources	Same qwen3.5:4b, supplied pool and BM25 top 3
Resource ceilings	4 retrievals and 18 LLM attempts including the final call
Answer policy	Same final synthesizer, scoring and post-inference gold
Uncertainty	Resample 100 question clusters with both seeds together

Actual cost is measured. A matched cap does not make token use equal.

Final benchmark results



+22.0 points

RAIR over earlier RAIR core
95% interval +15.0 to +29.5

+4.5 points

RAIR over KAIR
95% interval -3.0 to +12.0
Superiority is not established

200 runs per arm. Errors and abstentions remain in the denominator.

Resource use and answer coverage

Metric	Updated RAIR	KAIR
Exact match	40.5%	36.0%
Answer F1	47.1%	45.5%
Answered coverage	69.5%	73.5%
Mean LLM calls	12.27	9.57
Mean total tokens	9,657	11,364
Mean retrievals	2.69	2.13

RAIR uses more calls but fewer total tokens in this experiment.

Remaining outcomes

Outcome out of 200	RAIR	KAIR
Exact answer match	81	72
Answered but not exact	58	75
Abstained	61	50
Question execution error	0	3

RAIR abstentions

39 budget stops, 15 repeated queries
7 after planned gaps were resolved

Nonexact answers include both real errors and wording differences.

What the failed traces reveal

Bridge evidence can be rejected

RAIR discards the coaching-at-Fordham link because it is not a city.

Negative evidence can be lost

Welsh nationality is rejected, then a yes answer is inferred from one person.

Metrics can penalize wording

Both engines return 5 against five books, receiving zero token overlap.

Strict validation can retire agents

Duplicate document selections cause both KAIR proposals to fail.

Illustrative cases only. The benchmark scores are unchanged.

The working local dashboard

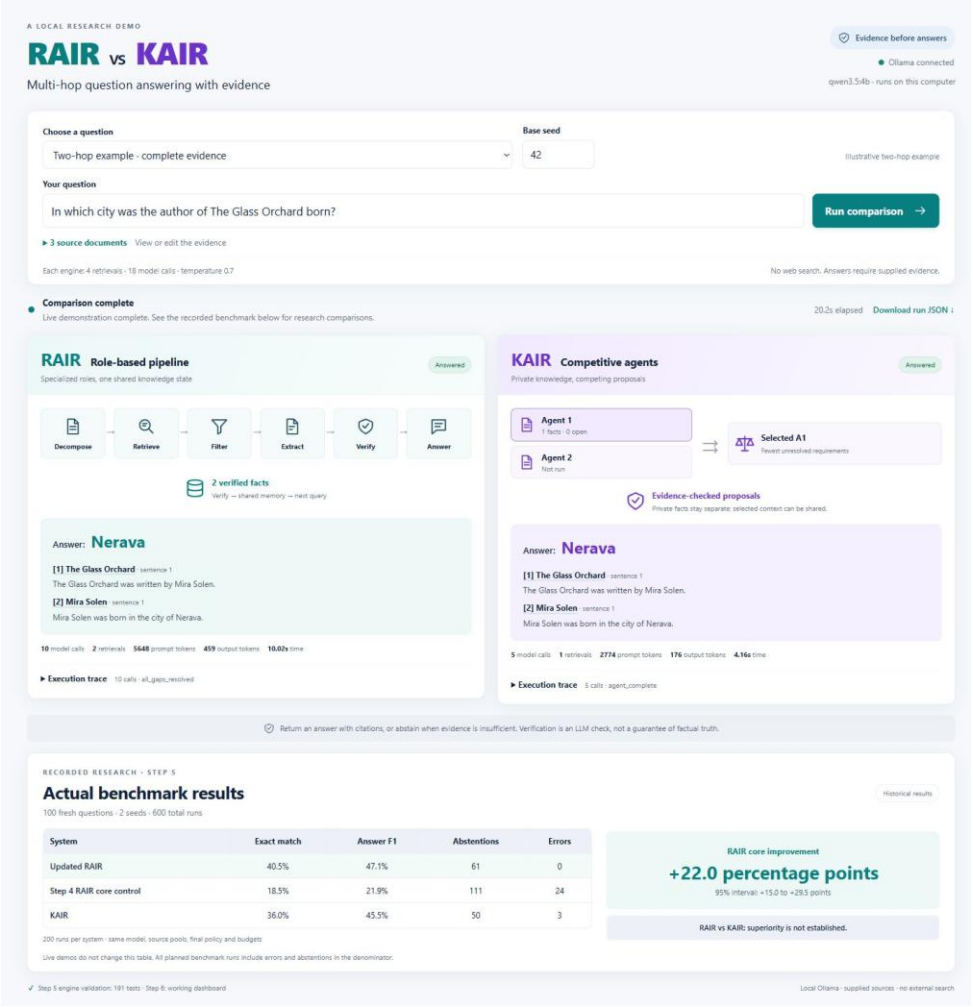
http://127.0.0.1:8765/

Inspect the evidence

Run the same question and sources through both actual controllers.

Explain the decision

Open source citations, gaps, agent proposals and raw model traces.



Verification before presentation

207 automated tests passed

Engine behavior, evaluation controls and dashboard contracts.

600 frozen runs rechecked

Scores, call budgets, source references and confidence intervals.

Real browser evidence retained

Complete, missing, custom and labeled development scenarios.

Conclusion and next research steps

A working and inspectable local QA comparison

Clear improvement over the earlier RAIR core.
The comparison with KAIR remains inconclusive.

Next research steps

Improve bridge and negative evidence handling, then test on a new untouched cohort with additional models and component ablations.